Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Indomie
Project / Solution Name	NORDYC
Name (Leader)	Nelsen Keane Gunawan
Email & Phone (Leader)	DMT2309281@xmu.edu.my & (+62) 859 2065 5352
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://nordyc-academic.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/RikoHalim/nordyc

S2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

NORDYC uses a cloud-based architecture that connects the user interface, backend, AI services, and database. The frontend is developed using React and TypeScript, providing an interactive dashboard where universities can upload curriculum documents, review course information, explore future career opportunities, identify competency gaps, and generate academic improvement plans.

The frontend communicates with a Python FastAPI backend through REST APIs. The backend manages user requests, document extraction, data validation, AI analysis, and academic recommendations. Background workers handle processing tasks to keep the application responsive.

For AI-powered analysis, NORDYC integrates Anthropic Claude Opus 5 through Replit AI Integrations. Brave Search API supports external research and workforce information discovery. Retrieved information is validated against approved sources before being used in academic analysis.

PostgreSQL stores curriculum records, workforce evidence, analysis results, recommendations, and review histories. SQLAlchemy manages database interactions, while Alembic handles database migrations.

The application uses Replit as its cloud development and intended hosting platform, connecting the frontend, backend, AI services, and database into one integrated system.

This architecture supports traceable academic recommendations and evidence-based curriculum improvement.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12.12	Backend programming language and AI processing	PSD License
TypeScript 5.9.3	Frontend programming and type safety	APache 2.0
React 19.3.1	Interactive user interface and dashboard	MIT License
FastAPI	REST APIs, document processing and AI Integratio	MIT Lincense
PostgreSQL 16.15	Database for curriculum data, analysis results, and recommendations	PostgreSQL License
Vite 8.3.5	Frontend development and production builds	MIT Licencse

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Anthropic Claude via Replit-managed AI	Generates structured, evidence-grounded curriculum findings, with schema and citation validation. Earlier live verification is documented; the current synthetic demo uses mocked responses.	**Paid — Replit credits.** Sponsor-provided access is not confirmed.
Brave Search API — Brave	Optional discovery of relevant workforce and academic sources. Retrieved material requires source and permission checks before admission as evidence. Disabled in the current demo.	**Separate provider plan; provisioned tier unconfirmed.**
Gemini API — Google	An implemented alternative AI provider with explicitly controlled fallback. Currently disabled; not claimed as verified live usage.	**Not currently provisioned; free/paid tier not established.**

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Cloud Platform	Cloud hosting environment for React frontend and FastAPI backend	Replit Development Preview / Region not verified
PostgreSQL 16.15	Stores curriculum records, AI analysis results, and academic recommendations	Replit test environment / Region not verified

Service & Provider	Purpose in Project	Plan / Region
Replit AI Integration	Provides managed access to Anthropic Claude Opus 5 for AI-powered curriculum Analysis	Managed AI service / Region not verified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude — recorded model identifier: claude-opus-5	Anthropic through Replit-managed AI	Pre-trained. Generates structured curriculum findings from supplied course information and evidence. No fine-tuning or self-training. Earlier live verification is documented.	Proprietary hosted service, subject to applicable Anthropic and Replit service terms; model weights are not distributed.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic programme, curriculum, evidence, and review fixtures	Project `examples/`, `fixtures/`, and synthetic verification scripts	Synthetic	Project-generated; no separate dataset licence declared. Genuine excerpts copied into synthetic verification lanes retain their original terms.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.0.3 occupation–skill subset	European Commission: https://ec.europa.eu/esco/download/	Public, legacy subset	ESCO reuse conditions under Decision 2011/833/EU; acknowledgement and adaptation notice required.	N
MQA programme-standard summaries: Computing, Accounting, Business Studies, Biotechnology	Malaysian Qualifications Agency: https://www2.mqa.gov.my/qad/v2/types3new.cfm	Public-source, reviewed summaries	Copyright retained. No general open licence established; only scoped, attributed summaries/short quotations are recorded.	N
XMUM reviewed programme/course inventory	Xiamen University Malaysia: https://xmu.edu.my/undergraduate-programmes	Public-source, manually reviewed subset	University copyright; automation and redistribution permissions remain unresolved.	N
Job Market Insights, Q1 and Q2 2026	Department of Statistics Malaysia: https://www.dosm.gov.my/	Public, aggregate statistical subsets	DOSM/Government Open Data terms with attribution and applicable exceptions; not assumed to be CC BY.	N
Employment by MSIC Sector and Sex	OpenDOSM: https://open.dosm.gov.my/data-catalogue/employment_sector	Public, aggregate statistics	CC BY 4.0.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Generative AI and Jobs: A Refined Global Index of Occupational Exposure	ILO: https://www.ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure	Public, reviewed evidence cards	CC BY 4.0, subject to third-party exclusions and adaptation requirements.	N
Artificial Intelligence and the Changing Demand for Skills in the Labour Market	OECD: https://doi.org/10.1787/88684e36-en	Public-source, reviewed summaries	OECD copyright and applicable publication terms; no open microdata licence verified.	N
Navigating the Jagged Technological Frontier	https://doi.org/10.1287/orsc.2025.21838	Public, reviewed findings—not participant-level data	CC BY 4.0, as recorded in the dataset registry.	N
METR developer-productivity study and follow-up	https://metr.org/blog/	Public-source, limited evidence cards	No reuse licence established in the inspected records; further rights review required.	N
Autonomous laboratory study and author correction	https://pmc.ncbi.nlm.nih.gov/articles/PMC10700133/ and https://pmc.ncbi.nlm.nih.gov/articles/PMC12872444/	Public, reviewed scientific evidence	CC BY 4.0, subject to applicable third-party exceptions.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
UN 2030 Agenda goals and targets, A/RES/70/1	https://digitallibrary.un.org/record/3923923	Public, official resolution	Recorded as a public-domain UN document with source acknowledgement; this does not extend to separate statistical API databases.	N
User-uploaded curriculum PDFs and reviewed inputs, if supplied	Institution/operator uploads	Private or user-supplied	Uploader must hold appropriate processing rights; no redistribution licence assumed.	**Possible**—depends on document contents.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit managed AI (anthropic)	Frontend/backend development, integration, debugging and testing	Major
OpenAI Chatgpt	System planning, architecture, UI/UX guidance and documentation	Major
OpenAI Codex	Code generation ,refactoring , debugging and automated testing	Major
ChatGPT image generation	Logo exploration, brandingconcept and visual design assets	Moderate
Replit-managed Claude Opus 5	AI integration development and validation	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team reviewed and tested AI-generated code, designs, and documentation before including them in NORDYC. We verified functionality through automated backend tests, PostgreSQL integration tests, frontend tests, and browser testing. AI-generated recommendations were checked for evidence accuracy, source citations, and uncertainty. We also reviewed code quality, security, and API integration, correcting errors when identified. AI tools supported our development process, but our team remained responsible for technical decisions, validation, and the final project.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud Development & Hosting Platform	Used to develop, test, and prepare NORDYC for online deployment.
Replit	Replit AI Integrations (Anthropic)	Provides managed access to Claude Opus 5 for AI-powered curriculum analysis.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was important for developing, integrating, and testing NORDYC. It provided a cloud development environment and access to AI integration services. However, Replit is not strictly essential to the system's core functionality. NORDYC could be hosted on other cloud platforms that support React, FastAPI, PostgreSQL, and background workers. Similarly, the AI integration could be configured through another compatible provider. Using Replit helped simplify development and integration, allowing our team to focus on building the application's main features.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The current synthetic demo uses fictional programmes and mocked AI responses; no document content is sent to AI APIs. When explicitly enabled, live analysis sends selected course information and permitted evidence excerpts to Replit-managed Anthropic Claude. Optional research uses generic queries rather than private documents.

Institution-scoped access controls restrict viewing and modification. Uploaded originals remain in restricted backend storage. Provider credentials stay server-side; operator credentials remain only in browser-tab memory.

There is currently no fixed retention period or comprehensive automatic deletion policy. Normal demo shutdown removes its temporary database schema and access credential, but uploaded originals and operational artifacts may remain. Retention schedules and deletion procedures must be established before an institutional pilot.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- Bias and incomplete evidence: Occupational benchmarks may be outdated or poorly representative of Malaysian disciplines. Outputs retain source dates, coverage gaps, and uncertainty; human review remains essential.
- Limited curriculum detail: Course titles do not establish taught competencies. Title-only findings remain preliminary and flag syllabus confirmation rather than definitive replacement decisions.
- Provider dependency and rate limits: Live analysis depends primarily on Replit-managed Claude. Timeouts, bounded attempts, and recorded failure states constrain execution. Gemini fallback requires explicit configuration and is disabled in the current demo.
- Unproven academic accuracy: Synthetic tests verify software behaviour, not recommendation quality. Independent expert evaluation is still required before institutional adoption.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Nelsen Keane Gunawan

24 October 2026